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THERMAL ANALYSIS OF THE NATURAL REMANENT MAGNETISM OF SOME UPPER SILURIAN RED SANDSTONES IN THE OSLO REGION

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SUMMARY

Thermal demagnetization studies of the Upper Silurian Ringerike Sandstone in the Oslo area, Norway, have revealed the presence of two components of magnetization. The dominant component appears to have been acquired in Permian times when the rocks, due to extensive igneous activity in this area, most likely suffered an increase in temperature. The stable high temperature component estimated is in accordance with other Old Red Sandstone results of southern Norway and with certain U.S.S.R. data. The discrepancy between the Ringerike results and those obtained from a similar analysis of the Anglo-Welsh Cuvette Sandstones is discussed, concluding that most likely the secondary component in the latter formation has not been completely eliminated.

INTRODUCTION

The Caledonian orogeny brought about considerable geographical changes in the western part of the Scandinavian land mass, resulting in the emergence of large areas of Norway. The occurrence of conglomerates and breccias in most Old Red Sandstone (O.R.S.) deposits of southern Norway supplies evidence of a very energetic erosion process in the time subsequent to elevation, and the presence of folds and faults in these formations has revealed that orogenic movements were still active during and after sedimentation. These movements are especially pronounced in the western and northern areas but are considerably weaker in the southeastern parts of the country.

The present paper deals with the magnetism of the Ringerike Sandstone, the age of which ranges between late Ludlovian and Early Downtonian, in the Oslo area. This rock complex has in general been only slightly disturbed by Caledonian orogenic movements.

The purpose of investigating this formation was firstly to determine a pole position based on palaeomagnetic data from a relatively undisturbed formation, and secondly a tectonic aspect, i.e., to discover whether the Ringerike Sandstone may serve as a key formation for studying certain possible tectonic movements which may have taken place during the history of other O.R.S. deposits of southern Norway.

Unfortunately it appeared that the original magnetization of the

Ringerike Sandstone had been nearly completely obliterated by remagnetization during the Permian igneous period of this area, and an original component is considered to have been successfully derived from three localities only. Nevertheless, the results obtained are of great palaeomagnetic importance, and they have made possible an interesting comparison with data obtained from thermal demagnetization of the Anglo-Welsh Cuvette Sandstones (Chamalaun and Creer, 1964), a formation which is also considered as having been remagnetized in Hercynian times (Chamalaun, 1964).

GEOLOGY AND SAMPLING

The Ringerike Sandstone is a fine-grained sandstone of continental origin, varying in colour from grey to brownish red. In this investigation we are concerned with the brown or red sandstones only, as the grey ones are too weakly magnetized for measurement on the magnetometers presently available in our laboratory.

The formation represents the end of a thick sedimentary sequence, the deposition of which was started in the Cambrian and was finished at the Silurian–Devonian border. The stratigraphy of the type area (Ringerike) has been dealt with by Kiær (1908,1911), Störmer (1954) and Whitaker (1964). The lower part of the formation is rich in red mudstones, while in the higher levels grey and greenish micaceous siltstones and sandstones predominate.

The stratigraphic thickness is about 1,000 m in the northern area, thinning out southwards, being only about 300 m in the south. Faunal evidence (Kiær, 1911,1931) indicates a late Ludlovian age in the lower part of the sequence and an early Downtonian age in the uppermost part.

As the Ringerike Sandstone consists of thick competent beds, the Caledonian orogeny affected this formation only slightly. The bedding plane is more or less horizontal as the fold limbs normally dip at an angle of less than 20°. Locally, however, the dip of the layers may amount to 40–50° The trend of the folds is west-southwest to east-northeast. The age of the folding is considered to be Lower Devonian.

Today very small portions of the sandstone are available for study as the major area, which was originally supposed to have been occupied by this formation, either became covered by extensive lava flows or was displaced by plutonic rocks in the beginning of the Permian era. Fig.1 shows the location of the present sandstone areas in relation to other rocks of the Oslo region. The whole rock complex of this region is preserved due to a downward faulting which is probably associated with the great displacement of magma.

Although there is also a large number of extensive Permian faults within the Oslo Graben, there are no indications that these movements have significantly influenced the attitude of the layers in comparison with that of the Caledonian folding. On the other hand the sandstone was obviously heated by the Permian lavas and intrusions. Thermometamorphism may be recognized at a distance of up to 2 km from the igneous contacts, and it is not unlikely that there was a high geothermal gradient during this time in the entire region.

From the areas of Ringerike, Kolsås, Sande and Sylling a total of

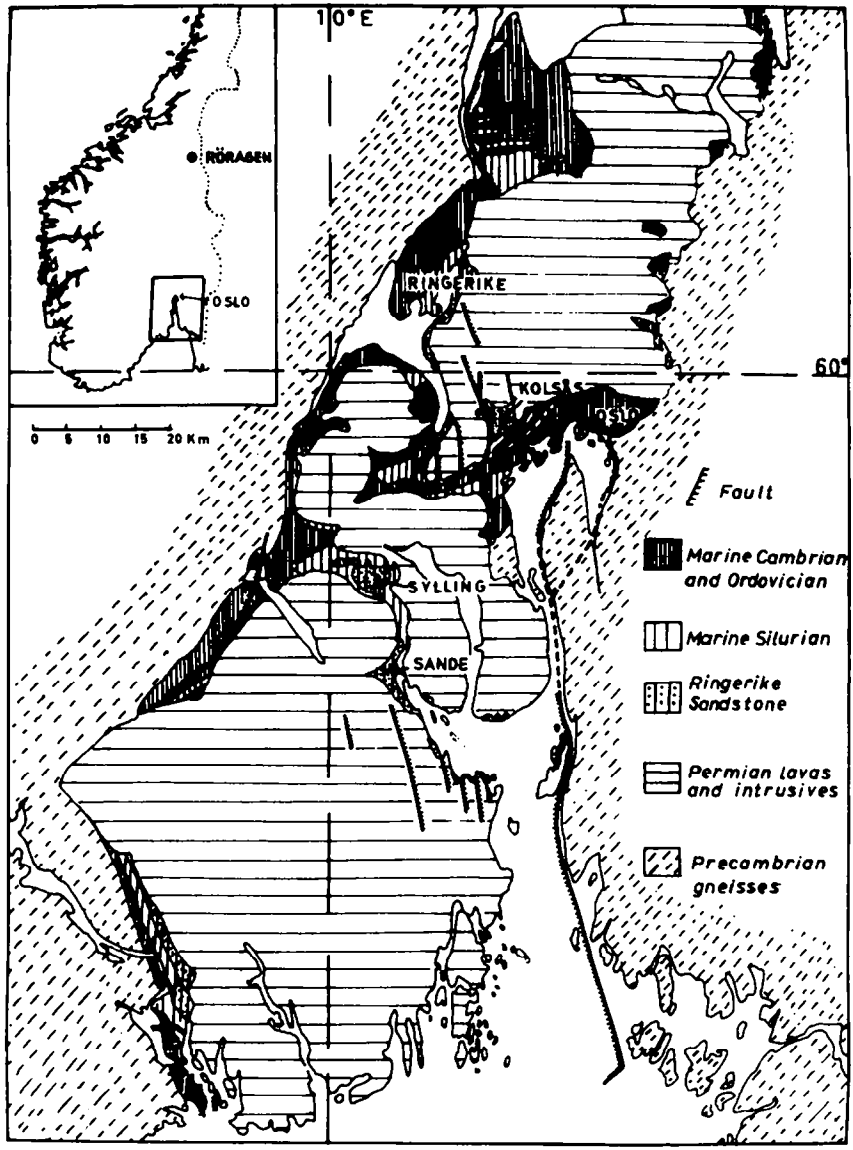


Fig.1. Geological sketch map of the Oslo area. Simplified after Holtedahl (1953, plate 14).

79 samples from 23 sites were collected. Ringerike is represented by 19 sites altogether as most red beds exposures are located in this area. Most samples were obtained from road cuttings.

EXPERIMENTAL METHOD

The remanent magnetism of the cylindrical specimens was measured on an astatic magnetometer, following the method described by Collinson et al. (1957). The remanent moment is determined to within 2° in direction and about 5% in intensity.

As thermal demagnetization of red sediments has proved very useful for analysing the magnetism of such rocks (Irving et al., 1961; Chamalaun and Creer, 1964; Irving and Opdyke, 1965), this experimental procedure was adopted in the present study. The heating of the specimens, twelve in each run, was performed in several steps to successively higher temperatures, followed by rapid air cooling to room temperature after each heating. The heating process was carried out in air and made as rapid as possible to prevent severe mineralogical changes. Before heating, the field in the space occupied by the specimens was reduced by means of three pairs of square coils and the field cancelled to 5γ or less within the volume occupied by the specimens. When cooling was finished, the field had generally increased to values of $10–60\gamma$.

In the first series of heatings the specimens, between demagnetization and measurement, were left in the earth's field for a maximum of 3 h. This had serious consequences for the results from weaker rocks obtained after heating above 500–600°C, because the rocks, due to heat treatment, often acquired a viscous magnetism of substantial magnitude compared with the remaining remanence. This difficulty was overcome when zero field storage became available.

The electrical furnace, which is housed in the magnetometer hut, has, as far as possible, been made of nonmagnetic materials. A detailed description of construction, similar to that used by Irving et al. (1961), and the heating technique, are given by Helle (1966).

THE NATURAL REMANENT MAGNETISM

The intensity of magnetization at the 23 sites investigated ranged between $3.5 \cdot 10^{-6}$ e.m.u./cm^3 and $29 \cdot 10^{-6}$ e.m.u./cm^3. The natural remanent magnetism (n.r.m.) is in general characterized by a strong agreement between cylinders from the same sample as well as between samples within the same site. In all cases the remanent directions show a strong divergence from that of the present geomagnetic field.

The site mean directions before and after geological tilt correction are illustrated in Fig.2 and 3. These figures show that the grouping is improved when dip corrections are neglected. The results before dip corrections and those obtained from the Permian igneous rocks of this area (Van Everdingen, 1960) are in excellent accord with each other. On the other hand the results, even when referred to the bedding plane, deviate from those obtained from the Lower Devonian Røragen Formation (Storetvedt and Gjellestad, 1966).

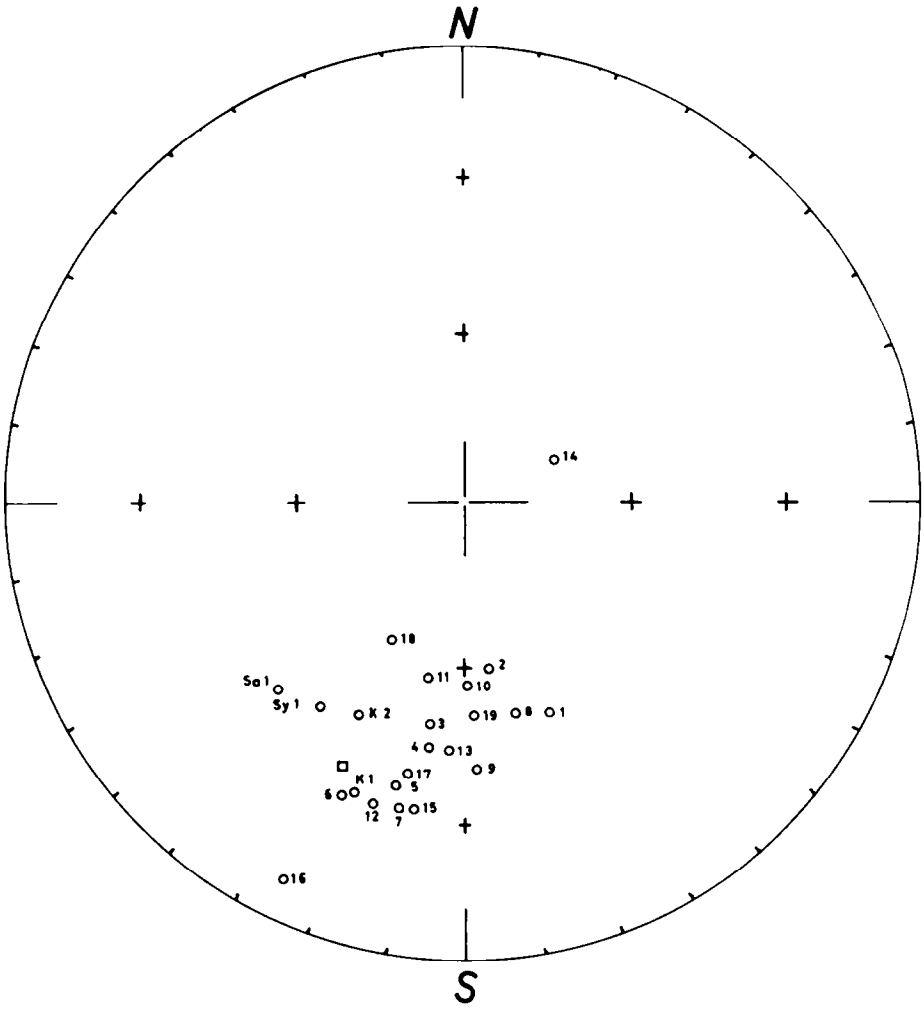


Fig.2. Equal area plots of site mean directions before correction for folding. Plots are northseeking magnetism in the upper hemisphere (negative). The square denotes the mean Permian direction estimated from the igneous rocks of the Oslo region.

Special attention has been paid to the Kolsås area where the sandstones are thermo-metamorphic nearly everywhere. Probably as a result of heating the samples from this area are very hard and in some cases they were impossible to core. Accordingly, a severe Permian remagnetization may be expected to have occurred, an assumption which has been confirmed by heating experiments.

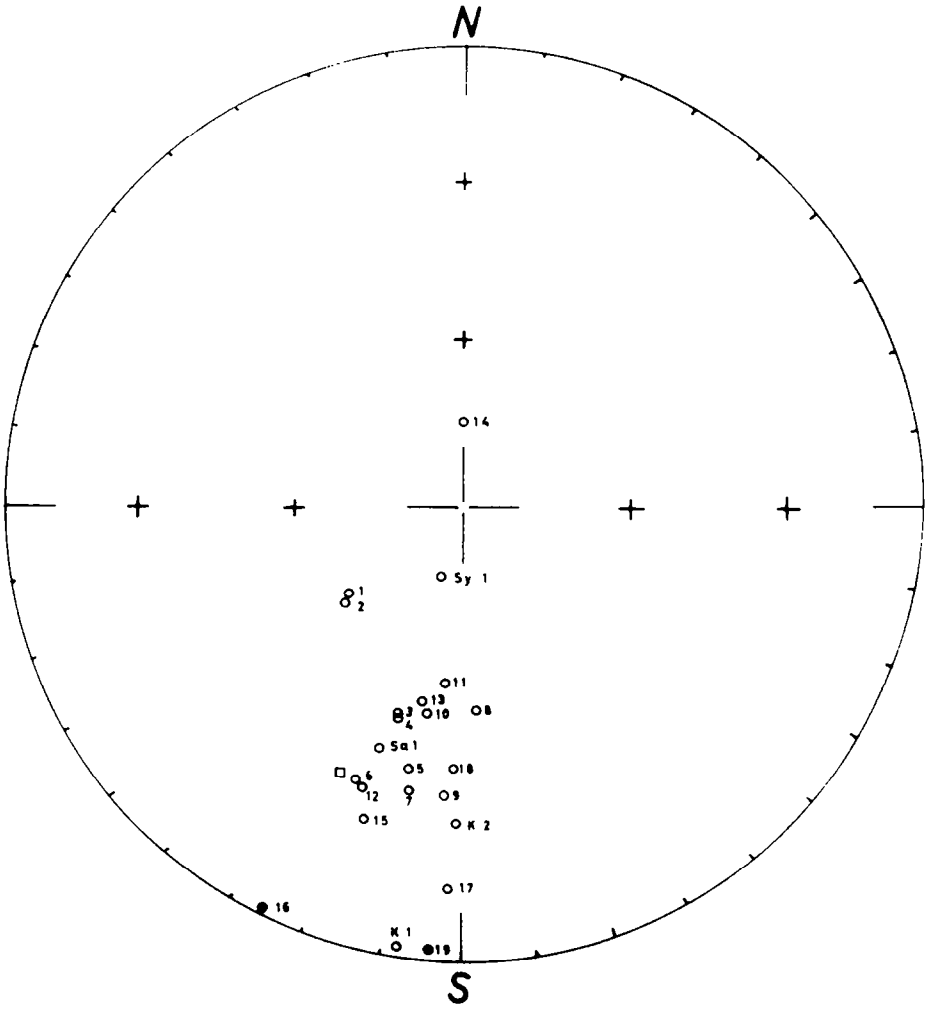


Fig.3. Site mean directions after geological tilt correction. Remarks as in Fig.2.

The bedding plane of the two sites collected in the Kolsås area (K1 and K2) dips at angles of 35° and 49° respectively. Fig.2 and 3 show that these two site directions are excellently grouped around the Permian field when referred to the present horizontal, but not when referred to the bedding plane. Thus, application of the fold test in this area suggests a folding prior to the igneous period, in agreement with geological observations which show that the folds are cut by the sub-Permian peneplane.

Although the n.r.m. results agree strongly with the Permian field

estimated for this region, a certain planar distribution is evident. As no low stability components appear to be present, this behavior of the remanent magnetism probably reflects a remaining component of pre-Permian origin.

THERMAL EXPERIMENTS

Thermal demagnetization was performed on specimens from a total of 21 samples taken from 10 different sites. The sites studied were the Ringerike (R) sites no.1, 3, 4, 8, 10, 11 and 14, the Kolsås (K) sites no.1 and 2 and the Sylling (Sy) site no.1.

The hypothesis of a composite nature of the natural remanent magnetism was easily verified, as a general and consistent directional change occurred during demagnetization in nearly all the samples investigated. The only exceptions are samples which were collected in the vicinity of igneous contacts. These samples showed a complete thermal stability of the n.r.m.-directions and it is assumed that they have become thoroughly remagnetized in the Permian field, the relevant agency in the remagnetization process being the heat from cooling igneous bodies.

A study of the thermal decay curves of the n.r.m. moment reveals the presence of two components, the moment decreasing rapidly in the temperature ranges 400–550°C and 650–700°C respectively. The relative importance of these two components may vary largely from site to site and sometimes from sample to sample at the same site. Both components appear to contribute to the n.r.m. in the majority of cases. In some samples the lower temperature range carries the essential part of the remanence, but in other cases nearly all magnetism is due to blocking temperatures around 650°C.

The lower temperature range corresponds to that of nearly pure magnetite and it is tentatively suggested therefore that this mineral is in part an important carrier of the n.r.m. of these red sandstones. On the other hand, it is fairly certain that the magnetic constituent above 600°C is haematite.

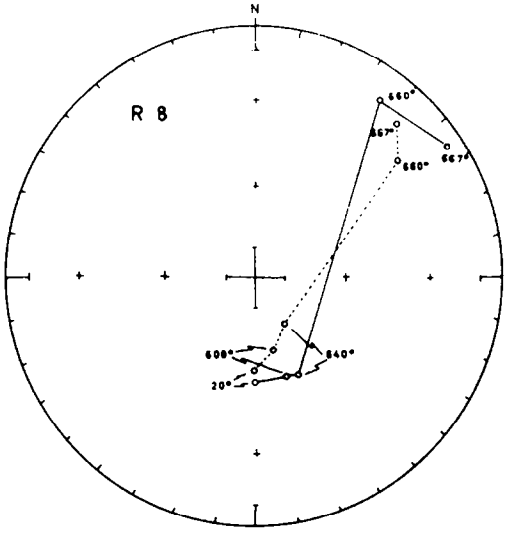
In general there are no essential directional or intensity changes below 400°C. In some cases the large changes of the remanent vector occur in the interval 400–600°C, and in other cases the major changes are concentrated around 650° C where the magnetic moment decays rapidly to zero.

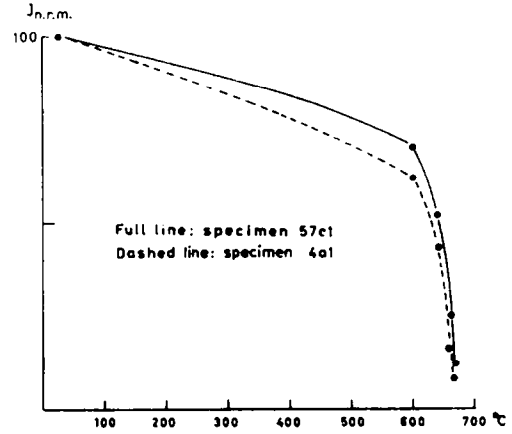
The moments change towards normal polarity with shallow negative inclinations, in most cases a total change of more than 100° of arc.

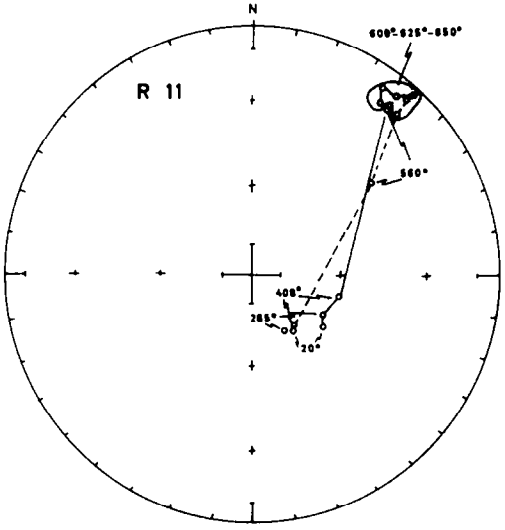
Stable directions do not exist below 600°C and they are in general difficult to estimate. The problem of detecting the stable directions is either that the remanence above 600°C is too weak for reliable measurement, or that the moment in question resides in grains having blocking temperatures too close to the Curie point of haematite to record with sufficient reliability the point at which the secondary component disappears.

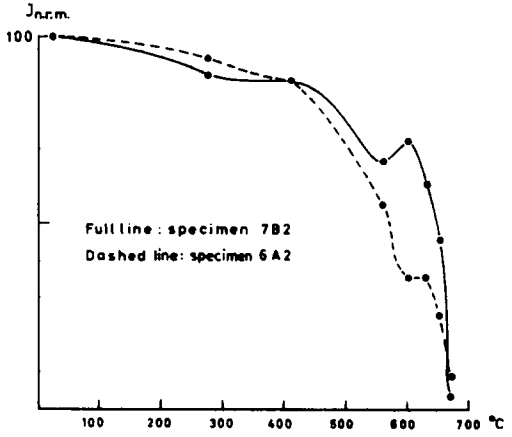
Fig.4 and 5 show typical behavior patterns obtained during demagnetization.

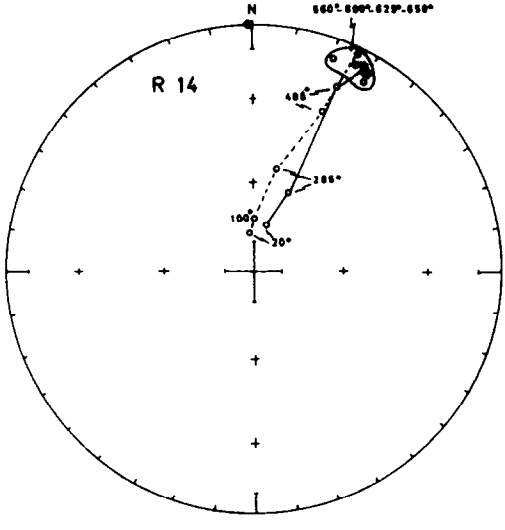
Samples from site R 14 (Fig.4) do not show an intensity decrease above 400°C but instead an intensity increase by about 30–40% occurs. As the secondary magnetization is not too far from being diametrically opposed to the

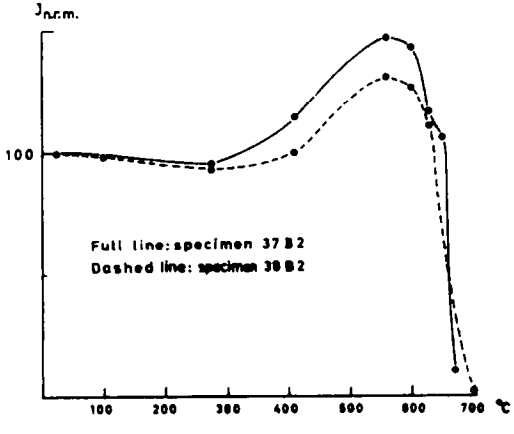

N
R 8
660°
667°
660°
600°
20°
640°


Jn.r.m.
100
Full line: specimen 57c1
Dashed line: specimen 4a1
100
200
300
400
500
600
700 °C


N
R 11
600°-625°-650°
560°
405°
285°
20°


Jn.r.m.
100
Full line : specimen 7B2
Dashed line: specimen 6A2
100
200
300
400
500
600
700 °C


N
560°-600°-625°-650°
R 14
405°
285°
100°
20°


Jn.r.m.
100
Full line: specimen 37 B2
Dashed line: specimen 38 B2
100
200
300
400
500
600
700 °C

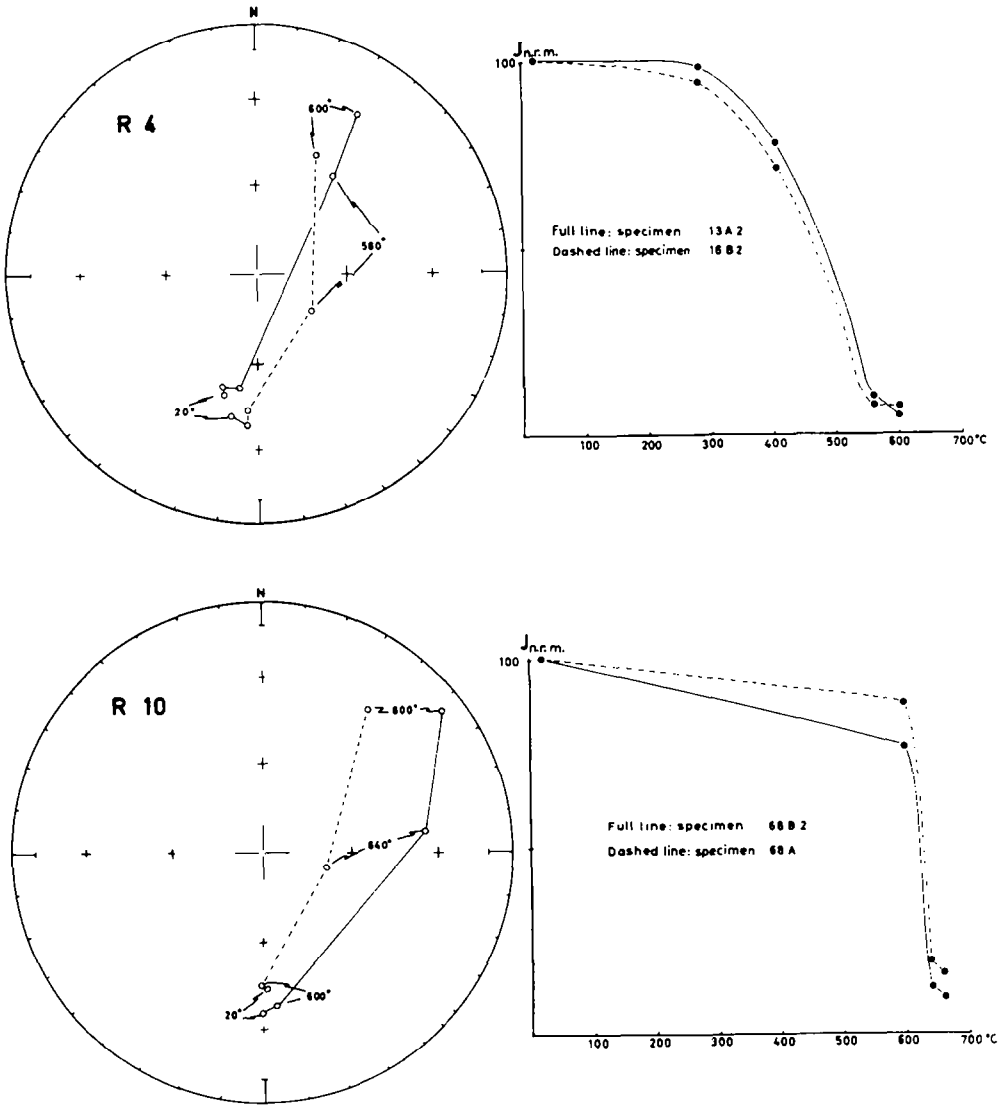


Fig.5. Direction and intensity of remanent magnetization versus temperature for specimens in which the stable moment has not been determined. Directions are referred to the bedding plane.

Fig.4. Directional and intensity changes (on equal area nets and normalized scales respectively) during demagnetization of specimens in which the high temperature component has been estimated. Directions are referred to the bedding plane.

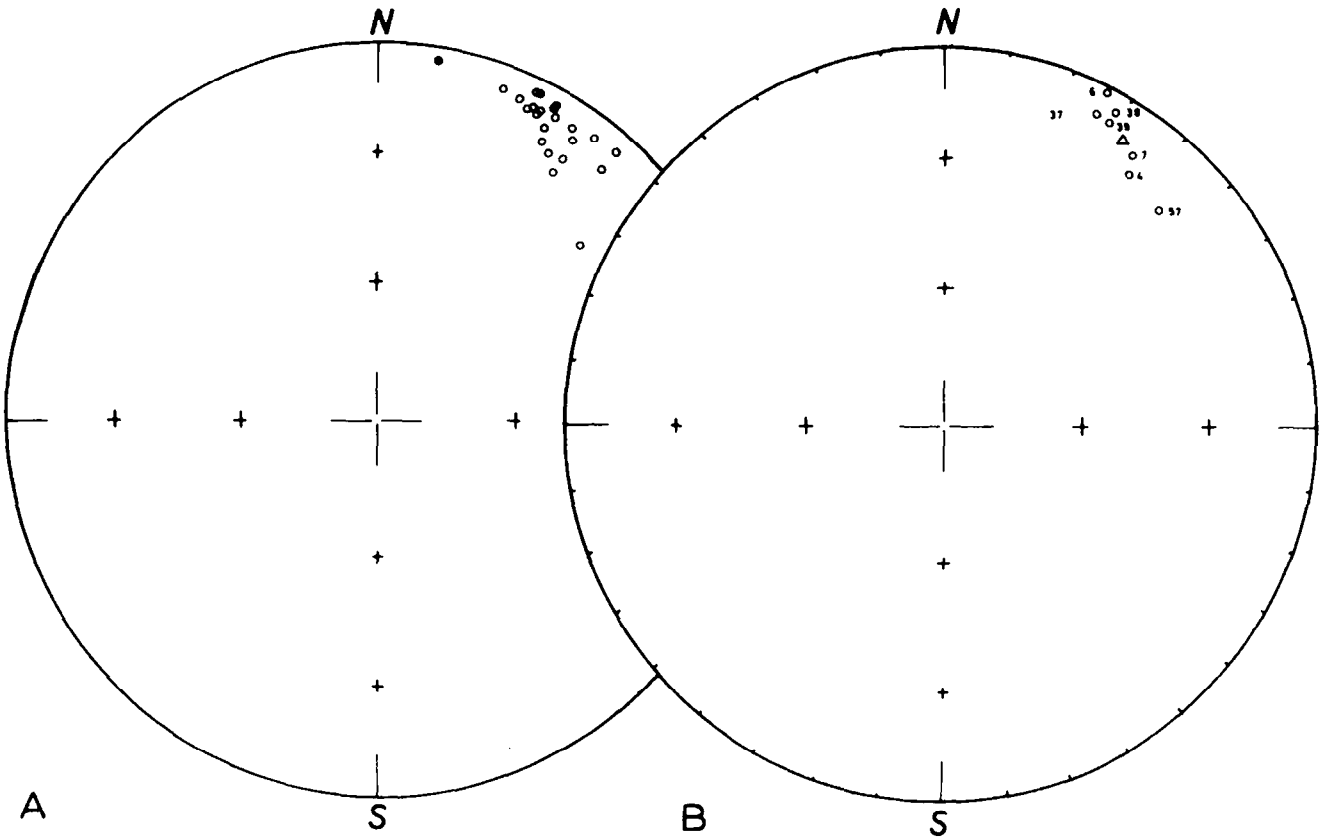


Fig.6. Stable high temperature directions referred to the bedding plane: A. all specimen directions; B. sample mean directions. The triangle represents the mean direction based on unit weight of each sample.

stable high temperature component, which is unusually strong in this locality, the intensity increase may be attributed to the demagnetization of a certain moment in the temperature range between 400°C and 600°C.

Artificial thermoremanent magnetization was imposed on specimens from R 4 and R 14. The thermal decay of this magnetization indicates the presence of two well defined components in both localities, the major part of the intensity being blocked below 600°C. The decay curves are nearly identical at the two sites but the haematite moment is considerably larger in R 14. These results are in complete agreement with the results of thermal demagnetization of the corresponding natural remanences (Fig.4 and 5).

The high temperature direction has so far been successfully estimated from seven samples only; one more than included in a previous report . (Storetvedt, 1967). The thermally stable directions represent the Ringerike (R) sites no.8, 11 and 14. Fig.6 shows all specimen directions as well as the different sample means. In R 11 and R 14 the specimen directions after thermal cleaning at 650°C were combined giving each unit weight to obtain the mean of each sample. For site R 8 the directions after cleaning at 667°C were used. The overall mean direction, giving each sample unit weight, has a declination of N 31°E and an inclination of 11° up (negative).

PALAEOMAGNETIC RESULTS

As the attitude of the layers at the sites considered is nearly horizontal, the fold test cannot be used for dating the high temperature magnetization relative to the age of the folding. For further palaeomagnetic conclusions therefore one has to assume that this magnetization direction was acquired by the rock at the time of its formation.

Apart from polarity, the mean direction of the thermally stable directions in the Ringerike formation is nearly identical with that of the Lower Devonian sediments at Røragen (N 210°E, + 9), the remanent direction of which has been proved to possess directional stability throughout the whole range of temperatures (Storetvedt and Gjellestad, 1966). Furthermore, palaeomagnetic results from a third O.R.S. formation, of southern Norway (on the west coast) have recently become available. A low stability component in this formation, carrying a secondary component aligned along the present field direction, is usually removed when demagnetization is affected at 400°C (Lie, 1967). The results obtained so far (from about 40 cleaned samples) are in accordance with those previously mentioned. It would seem justified therefore, despite the small amount of palaeomagnetic information, to calculate a pole position from the result of the Ringerike formation. Based on the mean direction (031°, -11°) and on the hypothesis of an axial dipole configuration of the earth's magnetic field, a pole location at the present geographical coordinates 159°E, 21°N is derived.

There is some evidence that the inclination in sedimentary rocks may exhibit systematic errors. A recent analysis (Van Andel and Hospers, 1966) suggests a general flattening of the inclination in all sedimentary rocks of all ages ranging from Cambrian to Quaternary. If such a flattening is real in red sediments, one would expect a latitude dependence, and it appears likely that the error would be negligible in equatorial regions. As

the Norwegian Old Red Sandstones are low latitude deposits and the different formations, despite the variation in depositional conditions and in lithology, contain consistent permanent magnetizations, there is a strong evidence that the results obtained have a real bearing on the attitude of the geomagnetic dipole axis at the time the rocks were formed.

COMPARISON WITH OTHER EUROPEAN AND WITH U.S.S.R. RESULTS

A study similar to that reported in this paper was carried out by Chamalaun and Creer (1964) on the Old Red Sandstones of southwest England. They claimed to have isolated a weak component of high thermal stability, which could only be detected by heating above 600°C. However, a study of their demagnetization trends (Chamalaun, 1964) does not reveal any evidence in support of this statement, as a systematic directional change is recorded up to 650°C, the last step in their demagnetization procedure at which the remanence could be studied. The reliability of their high temperature direction has, for the same reason, also been questioned by Van Hilten (1965). The fold test that was applied (which assumes that the high temperature direction was acquired prior to the Hercynian folding of that area) is also not conclusive evidence that all secondary magnetism has been removed.

The problem of estimating the final direction appears basically to be the same in both formations - the remagnetization process has affected the haematite component at blockage temperatures immediately below the Curie point of this mineral. For this reason it is inappropriate to apply heating steps of 50° as Chamalaun and Creer (1964) have done. In this context the reader must be referred to Fig.4 and 5 of the present paper where it is shown that substantial changes in the remanent magnetism may occur between 640° and 660°C.

The important difference in the results of the two formations is, therefore, that a final direction, which most likely represents the original magnetization, appears to have been successfully derived in the Ringerike Formation only. This stable direction is significantly different from that given by Chamalaun and Creer (1964).

As some of the results obtained by Stubbs (1958) on the Lower Devonian lavas of southern Scotland agree with the thermal demagnetization results from southwest England, Chamalaun and Creer concluded that the n.r.m.-directions carried by the lavas most likely reflect the true Devonian field for Europe. Nevertheless, serious doubt can also be thrown on the palaeomagnetic reliability of the n.r.m.-directions of the Scottish lavas. When Chamalaun and Creer (1964) proposed a revision of the Devonian pole, they probably did not take into consideration the possibility that the lavas might have become remagnetized at a later time as well. A more detailed discussion of this problem has recently been given (Storetvedt, 1967), concluding that the revised Devonian pole has most likely been based on inadequate evidence.

Several O.R.S. formations from elsewhere in Europe and the U.S.S.R. have been studied palaeomagnetically, but a thermal analysis of their magnetism is not known to have been performed. Many of the pole positions obtained do not differ significantly from those based on the Norwegian results,

but they are all systematically displaced towards slightly higher latitudes (in the present northern Pacific Ocean).

The reversal test in most of the U.S.S.R. formations does not seem convincing with respect to magnetic stability, as the two polarities show, even for large rock collections, a relatively large divergence from being exactly diametrically opposed. An exception seems to be the Lower Devonian Okler Series (Vlasov and Kovalenko, 1963) where the normal and reversed directions are nearly 180° apart. The pole position based on this formation is 163°E, 25°N, which strongly agrees with those calculated from the Norwegian data.

It seems pertinent to conclude therefore that a pole position around 160°E, 20°N is at present the best estimate of the Lower Devonian pole for Europe.

ACKNOWLEDGEMENTS

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